

ATPL EASA STUDY GUIDE | 062 - RADIO NAVIGATION

Chapter 10: Satellite Integrity Monitoring (RAIM & SBAS/GBAS)

Syllabus Source: CAE Oxford (pp. 385-424) | Pass Target: $\geq 90\%$ AviationExam

1. RECEIVER AUTONOMOUS INTEGRITY MONITORING (RAIM)

GPS satellites can transmit erroneous ranging signals without immediate warning. RAIM is an internal airborne algorithm that cross-checks satellite ranges for consistency:

- Fault Detection (FD): REQUIRES A MINIMUM OF 5 SATELLITES (or 4 satellites + Barometric VNAV aiding). Detects if a faulty satellite is providing corrupt data, alerting the pilot with an integrity warning.
- Fault Detection and Exclusion (FDE): REQUIRES A MINIMUM OF 6 SATELLITES (or 5 satellites + Baro-VNAV). Identifies WHICH satellite is faulty, excludes it from the navigation solution, and continues navigation without interruption!
- Mask Angle: Satellites within 5° of the horizon are rejected (masked) due to excessive tropospheric refraction.

2. SATELLITE-BASED AUGMENTATION SYSTEMS (SBAS)

SBAS uses ground reference networks and geostationary satellites to broadcast wide-area differential corrections:

- Systems: EGNOS (Europe), WAAS (USA), MSAS (Japan), GAGAN (India).
- Capability: Eliminates ionospheric delay errors. Enables LPV (Localizer Performance with Vertical Guidance) approaches down to ILS Cat I decision height (200 ft) without ground ILS equipment!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): RAIM Availability Requirements

Satellite Quantity for RAIM Functions: - Fault Detection (FD): Requires a minimum of 5 satellites visible (or 4 + baro-aiding). - Fault Detection & Exclusion (FDE): Requires a minimum of 6 satellites visible (or 5 + baro-aiding). Dispatch Rules: - If RAIM prediction predicts a loss of integrity > 15 minutes at ETA (± 15 min), flight cannot be dispatched solely on GNSS! - An alternate airport with conventional ground-based radio navigation (ILS/VOR) must be designated.

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What is the MINIMUM number of operational satellites required for RAIM to perform FAULT DETECTION (FDE)?

- [A]: 4 satellites
- [B]: AT LEAST 5 SATELLITES (or 4 with Barometric Altimeter aiding)
- [C]: 6 satellites
- [D]: 3 satellites

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Fault Detection requires 5 visible satellites with suitable geometry to detect an anomalous pseudo-range error.

Q2: How many satellites are required for RAIM to perform FAULT DETECTION AND EXCLUSION (FDE)?

- [A]: 5 satellites
- [B]: AT LEAST 6 SATELLITES (or 5 with Baro-VNAV aiding)
- [C]: 8 satellites
- [D]: 4 satellites

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. FDE requires 6 satellites: 4 to solve the 3D position/time, a 5th to detect the fault, and a 6th to isolate and exclude the corrupt satellite.

Q3: What is the European Satellite-Based Augmentation System (SBAS) called?

- [A]: WAAS
- [B]: EGNOS (European Geostationary Navigation Overlay Service)
- [C]: Galileo
- [D]: GLONASS

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. EGNOS is the European SBAS system that augments GPS signals to provide safety-of-life LPV approach capabilities.